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Pashto Poetry Attribution using Deep Learning Techniques

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***Abstract*—Pashto poetry, a rich tradition dating back to the 8th century, has been a cornerstone of cultural and literary heritage in the region. However, Pashto remains a low-resource language in computational linguistics, with limited annotated data and natural language processing tools. This scarcity poses challenges in preserving and promoting Pashto literature globally, sometimes leading to ambiguous poetry attribution due to oral traditions and lack of proper documentation. This study introduces the first curated Pashto poetry dataset, comprising 35 poets and 35,637 couplets, addressing the scarcity of digital resources. We employ machine learning and deep learning models for poet attribution, and propose a tailored data augmentation pipeline that substantially improves macro F1 score and accuracy. Random Forest achieves the highest accuracy of 85.25%. Our results demonstrate the efficacy of curated data and augmentation in advancing Pashto poetry classification, offering a foundation for future low-resource language research.**

I. Introduction

Poetry has been a cornerstone of cultural and literary heritage across the globe. Pashto, spoken by 50% of the Afghan population and 15% of the Pakistani population [1] [2], has a rich poetic tradition dating back to the 8th century with works of Amir Kror Suri [3]. Notable poets like Khushal Khan Khattak, Rahman Baba, Ghani Khan, and Hamza Baba have significantly contributed to this tradition. However, Pashto remains a low-resource language in computational linguistics, with limited annotated data and natural language processing (NLP) tools. This scarcity hinders the preservation and global promotion of Pashto literature, often resulting in unclbear poetry attribution due to oral traditions and inadequate documentation.

To address these challenges, we curated the first Pashto poetry dataset, comprising 35,637 couplets from 35 poets, collected from diverse sources including online blogs (e.g., Hamari Web, Rekhta), social media images processed via OCR, mobile apps, and scanned poetry books. Data scarcity was further mitigated through augmentation techniques, such as Random Swap and tailored strategies based on per-poet F1-scores. This paper employs machine learning (e.g., Random Forest) and deep learning (e.g., BiLSTM, transformers) models to classify Pashto poetry and attribute it to specific poets. Unlike prior work on languages like English, Hindi, and Persian, Pashto poetry classification remains underexplored, making our dataset and methodology significant contributions to low-resource NLP.

II. Related Work

Poet attribution using deep learning and machine learning has been explored across various languages. A study on Punjabi poetry [4] utilized a dataset of 830 poems from 11 poets, working across three scripts: Gurmukhi, Shahmukhi, and Roman. They employed various models including Bi-LSTM, Bi-GRU, DistilBERT, Softmax regression, SVM, and Random Forest. DistilBERT achieved the highest accuracy for Gurmukhi (90.36%) and Shahmukhi (87.95%) scripts, while Bi-LSTM performed best for the Roman script with 91.57% accuracy.

For Urdu Ghazals, researchers [5] worked with 18,609 couplets from 15 notable poets spanning the 18th-20th centuries. They experimented with traditional machine learning models (Logistic Regression, Random Forest, Naive Bayes, SVM) and deep learning models (LSTM, GRU, 1D-CNN). Transformer models showed superior performance, with BERT and RoBERTa achieving 80.32% and 79.52% accuracy respectively. Among traditional approaches, SVM and Logistic Regression achieved 64% and 60% accuracy, while LSTM performed best among deep learning methods at 59.96%.

Another study on Urdu Ghazals [6] focused on poetry generation using 17,609 couplets from 15 poets. They compared n-gram probabilistic models, LSTM, GRU, and GPT2. The n-gram model achieved the highest BLEU score of 0.5, followed closely by GPT2 with 0.48 for Urdu face and 0.6

for Urdu Awaz. The LSTM and GRU models showed lower performance with accuracies of 28% and 32% respectively.

Persian poetry classification [7] utilized 496 Persian Ghazals, focusing on era-based categorization. Their comparison of various models showed LSTM achieving the highest accuracy of 77.8% and F1-score of 76.6%. For Gujarati poetry [8], researchers created the "Kavan" dataset containing over 300 poems representing different "Rasa" emotions, achieving 87.62% accuracy with their deep learning classifier.

Several studies focused specifically on Urdu poet attribution. One study [9] using 1,563 poems from 5 poets found SVM performing best with 88.7% accuracy, followed by Naive Bayes at 84%. Another research [10] utilizing 11,406 couplets from 3 poets compared various models, with SVM achieving the highest accuracy of 82.85%. A third study [11] worked with 3,967 couplets from 4 poets, achieving 72% accuracy with SVM.

Arabic poetry attribution [12] utilized a substantial dataset of 18,646 Qasidahs from 73 poets, totaling over 1.8 million words. Their experiments with SVM and Naive Bayes both achieved remarkably high accuracy of 98.86%.

While not specifically focused on poetry, a study on Pashto text classification [13] demonstrated the potential for applying machine learning to Pashto text. Using various classification approaches, they achieved 94% accuracy with MLP classification, while a multilingual DistilBERT model, despite not being pretrained on Pashto, achieved 66.31% accuracy, suggesting potential for improvement with Pashto-specific models.

While work on Urdu, Persian, and Arabic poetry and text classification provides useful baselines, Pashto poses distinct challenges. Compared to Urdu and Persian, Pashto has fewer digitized corpora and limited pre-trained models; its orthographic and dialectal variation is less standardized than Arabic. These differences motivate our contribution: a dedicated, curated Pashto poetry dataset and experiments that evaluate how standard ML and transformer approaches perform in a truly low-resource setting.

III. Methodology

A. Dataset

The absence of an existing Pashto poetry dataset required a labor-intensive, multi-stage data collection process—one of the key contributions of this work. Pashto's limited digital presence and poor documentation posed major challenges, as most poetry was scattered across fragmented or inaccessible sources. We began with an extensive web search across blogs, articles, and platforms such as Hamari Web, WordPress, Rekhta, and social media, where couplets often appeared as images. However, the collected data remained insufficient, as many sources contained translations or incomplete couplets.

To expand the corpus, we collected poetry from prominent poets (e.g., Rahman Baba and Khushal Khan Khattak) via mobile and web applications. Due to text copying restrictions, we compiled hundreds of screenshots for Optical Character Recognition (OCR) processing and subsequent manual cleaning. When digital sources were insufficient, we purchased rare volumes from local libraries, manually scanned each page, and performed OCR conversion. This multi-stage, labor-intensive process established the dataset's value as a pioneering resource.

The dataset underwent a two-phase cleaning process, as shown in Figure 1. Manual cleaning first removed non-poetic content, such as prefaces and indices, followed by scripts that eliminated English text, symbols, and duplicates. Preprocessing focused on manual proofreading to minimize inconsistencies from mixing diverse sources and OCR errors. Although minor noise persists due to heterogeneous font styles and low-resolution images, this limitation is mitigated by the robustness of the deployed models. Due to resource constraints and the specialized nature of the poetry, we did not utilize external annotators for Inter-Annotator Agreement (IAA) or systematic quality assessment. Instead, internal validation was conducted via cross-referencing multiple sources and manual transcription reviews to ensure accurate poet attribution. Crucially, to prevent data leakage, couplets were split by poet rather than by couplet, ensuring that each poet's work appeared exclusively in only one data subset. The final dataset's reliability and usability are empirically substantiated by the resulting high F1-scores.

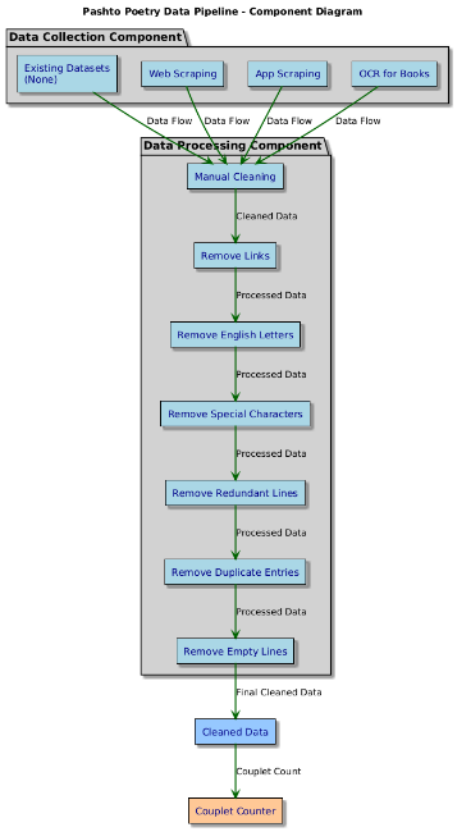


Figure 1: Data Cleaning Pipeline

The final dataset comprises 35,637 couplets from 35 poets, representing both historical figures (e.g., Rahman Baba, Khushal Khan Khattak, Hamza Baba) and contemporary poets (e.g., Ghani Khan, Khatir Afridi). This diversity ensures variation in styles, themes, word usage, dialects, and vocabulary, contributing to model robustness. Table I presents the distribution of couplets across poets.

The table below shows the number of couplets collected for each poet.

No.	Poet/Shayari	Couplets Count
1.	Abbasin Yousuf	1875
2.	Abid Malnak	534
3.	Ajmal Khattak	780
4.	Allama Abdul Hai	2380
5.	Aziz Mazerwal	30
6.	Bakht Zadah	565
7.	Da Ali Khan	977
8.	Ghani Khan	1399
9.	Gul Mansoor	487
10.	Hamza Baba	1177
11.	Haseena Gul	617
12.	Hazbullah Shahbaz	1216
13.	Javed Ahmedzai	561
14.	Jawad Azad	650
15.	Karan Khan	979
16.	Khaliq Ziari	40
17.	Khatir Afridi	1436
18.	Khayyam Yousufzai	533
19.	Khushal Khan Khattak	4257
20.	Matiullah Turab	1192
21.	Mohammad Ajanyar	672
22.	Mumtaz Orakazi	1645
23.	Munir Hunaray	471
24.	Munir Jan	1169
25.	Naeem Ahmed	856
26.	Rabia Mumtaz	572
27.	Rahman Baba	2202
28.	Rehmat Shah	1534
29.	Sahib Shah Sabir	1051
30.	Salim Riaz	232
31.	Shabbir Khan Durrani	1056
32.	Shakir Orakzai	569
33.	Shoaib Khan Khattak	615
34.	Suleman Shahbaz	655
35.	Zia Ur Rehman	653
Total		**35,637**

Table I. Poets and their couplets count

For the training of the models, we decided to exclude the poets that had couplet counts less than 500 (i.e., Aziz Mazerwal, Gul Mansoor, Khaliq Ziari, Munir Hanaray, and Salim Riaz) to ensure that the models had enough data to learn from. This resulted in a final dataset of 34,377 couplets from 30 poets where during this approach. For the remaining underrepresented poets with relatively fewer couplets, we used controlled synthetic generation to reduce bias and enhance the model's generalization capability. The dataset is publicly available on Hugging Face and can be accessed at https://huggingface.co/datasets/AliMuhammad73/Pashto-Poetry

IV. Experimentation and Results

A. Setup

Experiments were conducted on an 8-core Intel Core i7-12700 CPU (4.90 GHz, 64GB RAM) for machine learning (ML) models and two Tesla T4 GPUs with 32GB RAM on Kaggle for deep learning and transformer models. The dataset was split into 80% training and 20% testing for ML models, and 80% training, 10% validation, and 10% testing for deep learning and transformer models. Evaluation metrics included accuracy, and F1-score. For ML models, only accuracy was used. Our approach is benchmarked against established methods in text classification: Logistic Regression, SVM, and Random Forest. We deliberately focused on models that are accessible and interpretable in a low-resource context. Given the absence of existing public Pashto poetry attribution benchmarks, our models serve as the first robust baseline comparison for future research in this specific domain.

The ML pipeline involved label encoding, Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction (10,000 vocabulary size), and model training (Figure 2). The deep learning pipeline included one-hot encoding, tokenization (for transformers) or TF-IDF reshaping (5,000 vocabulary size for BiLSTM), embedding, and model training (Figure 3).



Figure 2: Machine Learning Pipeline



Figure 3: Deep Learning Pipeline

B. Model Configurations

We evaluated four ML models (Random Forest, Logistic Regression, Support Vector Machine, XGBoost), one deep learning model (BiLSTM), and seven transformer models (MT0 Base, DistilBERT, XLM-RoBERTa XL, Llama-3.2-1B, Small 100, MT5 Base, BERT-Based Multilingual Uncased). Extensive hyperparameter experimentation was conducted to optimize performance.

- **Random Forest**: Tested 50–200 estimators, max depths (None, 10–50). Final: 100 estimators, max depth=None.
- **Logistic Regression**: Varied C (0.1–100), solvers ('newton-cg', 'lbfgs'), max iterations (500–2000). Final: C=10, 'newton-cg', 1000 iterations.
- **Support Vector Machine**: Tried C (0.1–10), kernels ('rbf', 'linear'), gamma ('scale', 'auto'). Final: C=1.0, 'rbf', gamma='scale'.

- **XGBoost**: Experimented with 50–200 estimators, learning rates (0.01–0.3). Final: 100 estimators, default parameters.
- **BiLSTM**: Tested 1–2 bidirectional layers (64, 128, 256 units/direction), optimizers (Adam, AdamW, SGD), learning rates (0.0001–0.01), weight decays (0, 1e-4, 1e-5), early stopping patience (3–10). Final: 1 bidirectional layer (256 units/direction, 10.78M parameters), AdamW (learning rate=0.001, weight decay=1e-4), sparse categorical crossentropy, early stopping (patience=5).
- **Transformer Models**: Varied epochs (50–150), batch sizes (8, 16, 32, 64), learning rates (0.00001–0.001), weight decays (0, 1e-4), early stopping patience (3–10). Final: 100 epochs, batch size=32, AdamW (learning rate=0.001, weight decay=1e-4), cross-entropy loss, early stopping (patience=5).

C. Initial Experiments

Initial experiments trained all models on the unaugmented dataset. ML models utilized TF-IDF features, BiLSTM employed reshaped TF-IDF vectors, and transformers employed model-specific tokenizers. Results (Table II) showed Logistic Regression leading (62.88% accuracy), followed by Support Vector Machine (61.92%) and BiLSTM (59.63%). Transformer models underperformed (e.g., MT0 Base at 50.24%), mainly because they were not sufficiently pre-trained on Pashto text to capture the language's complex sub-word structures. Additionally, models tend to overfit on poets with high couplet counts (e.g., Khushal Khan Khattak), resulting in poor generalization for underrepresented poets.

Table II. Initial Results (No Augmentation)

Model	Accuracy (%)
Logistic Regression	62.88
Support Vector Machine	61.92
BiLSTM	59.63
Random Forest	51.93
MT0 Base	50.24
BERT Multilingual	49.25
Llama-3.2-1B	48.88
DistilBERT	45.39
MT5 Base	44.40
XGBoost	41.67

D. Data Augmentation

To address overfitting and improve generalization, we conducted three augmentation phases. First, we applied 25% augmentation (Random Swap, Random Deletion, Synonym Replacement) to DistilBERT and MT0 Base to identify effective techniques (Table III). Random Swap yielded the highest improvements (e.g., MT0 Base accuracy from 50.24% to 56.02%, DistilBERT from 42.63% to 44.81%).

In addition, Random Swap was also applied to Random Forest, for which the accuracy increased from 51.93% to 63.90%. Thus, we applied 50% Random Swap to Random Forest and BiLSTM. This improved their accuracy to 73.31% for Random Forest, and 66.13% for BiLSTM. A deeper, per-poet F1-score analysis showed significant gains, e.g., Shoaib Khan Khattak's F1-score rose from 0.1162 to 0.6846 (489.16% increase), which prompted us to implement tailored augmentation based on 50% Random Swap F1-scores:

Table III. Augmentation Results for DistilBERT and MT0 Base

Model	Accuracy (%)	F1 Score
DistilBERT (No Aug)	42.63	0.4109
DistilBERT (Random Deletion)	44.93	0.4378
DistilBERT (Random Swap)	44.81	0.4415
DistilBERT (Synonym Replacement)	43.13	0.4171
MT0 Base (No Aug)	50.24	–
MT0 Base (Random Deletion)	54.40	0.5384
MT0 Base (Random Swap)	56.02	0.5547
MT0 Base (Synonym Replacement)	52.87	0.5215

- Poets with F1 $>$ 0.76: 50% Random Swap.
- Poets with F1 between 0.7 and 0.76: 100% Random Swap.
- Poets with F1 $<$ 0.7: 100% Random Swap + 50% Random Deletion.

This expanded the dataset to more than 60,000 couplets, enhancing performance for low-couplet poets.

E. Final Results

Post-tailored augmentation, Random Forest achieved the highest accuracy (85.25%), followed by BiLSTM (82.73%) and Logistic Regression (76.66%) (Table IV). Transformer models improved modestly (e.g., MT0 Base at 67.20%), limited by Pashto pre-training. The tailored augmentation significantly boosted underrepresented poets' performance, e.g., Zia Ur Rehman's F1-score increased 289.06% from 0.1499 to 0.5832.

Table IV. Results with Tailored Augmentation

Model	Accuracy (%)
Random Forest	**85.25**
BiLSTM	82.73
Logistic Regression	76.66
Support Vector Machine	67.33
MT0 Base	67.20
XLM-RoBERTa XL	59.65

V. Discussion and Analysis

This study evaluated poet classification on a dataset of 35,637 Pashto couplets from 35 poets, enhanced by data augmentation to address class imbalance and data scarcity. Initially, without augmentation, Logistic Regression led with 62.88% accuracy, outperforming BiLSTM (59.63%) and transformer models (e.g., MT0 Base at 50.24%) due to effective TF-IDF feature extraction. However, post-augmentation, Random Forest achieved the highest accuracy (85.25%), followed by BiLSTM (82.73%), demonstrating the efficacy of tailored augmentation strategies.

Data augmentation significantly improved performance, particularly for poets with low initial F1-scores. The 25%

augmentation trials identified Random Swap as the most effective method, prompting a 50% Random Swap phase. The final tailored augmentation, based on F1-scores ($<$ 0.7: 100% Random Swap + 50% Random Deletion; 0.7-0.76: 100% Random Swap; $>$ 0.76: 50% Random Swap), expanded the dataset significantly, bossting F1-scores for underrepresented poets, e.g., Shoaib Khan Khattak's F1-score rose from 0.1162 to 0.6846 (489.16% increase) and Zia Ur Rehman's from 0.1499 to 0.5832 (289.06%). While Random Swap and Random Deletion improved model performance, we acknowledge that these operations may occasionally disrupt rhyme or meter patterns (known as wazan in Pashto poetry).

Historical poets (e.g., Khushal Khan Khattak, Rahman Baba) were consistently classified with high accuracy due to distinct stylistic features, while newer poets (e.g., Shoaib Khan Khattak, Zia Ur Rehman) posed challenges due to modern, overlapping styles. Random Forest's robustness to augmented data and BiLSTM's sequential modeling excelled, but transformer models (e.g., XLM-RoBERTa XL at 59.65%) underperformed, likely due to limited pre-training on Pashto.

A critical limitation of the current work is the absence of a formal ablation study to systematically quantify the contribution of key components. Future work will include a detailed ablation analysis, specifically measuring the performance drop when removing components like the tailored augmentation module and comparing the performance gains between TF-IDF features and new Pashto word embeddings. This systematic evaluation will establish which features contribute most significantly to the final attribution accuracy.

Limitations also include reliance on TF-IDF for machine learning and BiLSTM, which may miss semantic nuances; word embeddings could enhance performance. The large BiLSTM parameter count (10.78M) risks overfitting, though mitigated by early stopping. Transformer models require Pashto-specific pre-training for better results. The dataset, despite augmentation, may still lack sufficient couplets for some poets, and augmentation methods like back-translation were computationally constrained.

VI. Future Work

This pioneering effort to classify Pashto poetry, using a newly curated dataset faces several limitations. The dataset, while diverse, is limited for underrepresented poets (e.g., Aziz Mazerwal with only 30 couplets), exacerbating class imbalance and hindering generalization. Data collection via OCR from screenshots and scanned books introduced errors, necessitating labor-intensive manual cleaning. Reliance on TF-IDF features (10,000 for ML, 5,000 for BiLSTM) may miss semantic nuances, particularly for modern poets with similar styles. The BiLSTM's large parameter count (10.78M) risks overfitting, despite early stopping. Transformer models (e.g., MT0 Base at 67.20% accuracy) underperformed due to limited Pashto pre-training. While tailored augmentation (Random Swap, Random Deletion) improved performance, back-translation and synonym replacement were computationally costly and less effective, and augmentation risked introducing noise or semantic shifts, particularly with synonym replacement, which often fails to maintain the intricate stylistic and semantic constraints inherent in poetry. Finally, evaluation focused on accuracy and F1-scores, lacking deeper stylistic analysis to explain misclassifications.

While we report single-run accuracy and F1 scores, a limitation is the absence of formal statistical tests or confidence intervals. Future work will incorporate multiple runs and significance testing to ensure the reported performance metrics are statistically robust.

To address these limitations, future work will enhance the Pashto poet classification framework. Developing Pashto-specific word embeddings (e.g., FastText) could capture semantic nuances, improving differentiation of modern poets' styles over TF-IDF. Expanding the dataset with additional couplets, especially for underrepresented poets, will mitigate class imbalance. Advanced augmentation techniques, such as optimized back-translation or generative text models, could further diversify the dataset while minimizing noise. Pre-training or fine-tuning transformer models on Pashto corpora may boost their performance. Optimizing BiLSTM with dropout layers and exploring lightweight architectures will reduce overfitting. Additionally, analyzing misclassified poets' stylistic features (e.g., dialect, rhyme) could uncover factors affecting classification, strengthening the framework for low-resource NLP.

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